

PAVS: Bridging AMD's Softpower Gap

From the Voice AI Perspective

Why the best inference silicon doesn't win by itself — Voice AI as the lens, with the same pattern repeating in robotics, industrial & healthcare

The Voice AI: Business, Opportunities & Tech-Stack

A Massive Inference Market, Being Decided Now, Next is edge-inference

The opportunity

Metric	Value
AI voice-agents market, 2034	\$47.5B @ 35% CAGR
Cost per call — human vs AI	\$7–12 → ~\$0.40
YoY growth, production deploys	340% (500+ orgs)
Orgs planning voice AI by end-2026	80%

- A deployed agent is a **steady-state inference machine** — it doesn't train; it **transcribes, reasons & speaks 24x7**
- At scale, its **dominant operating cost is GPU compute**

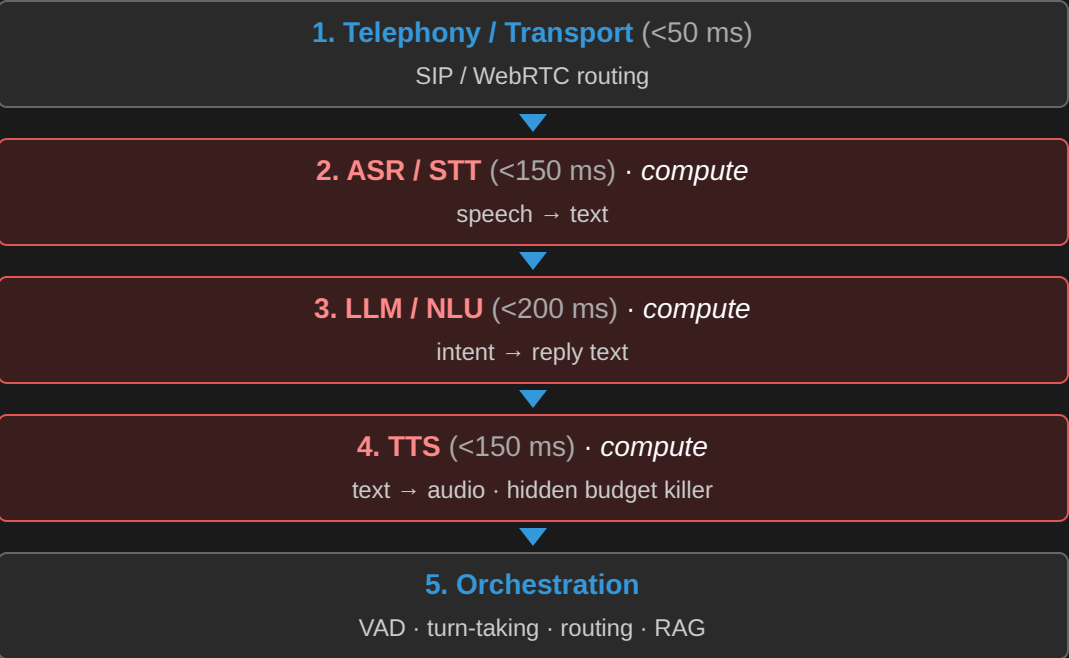
Why this is AMD's business

- **MI300X: 34% lower cost-per-token** for 70B-class LLM inference vs H100
- **Single-GPU fit** — 192 GB HBM3 runs a 70B model that needs **two** H100s
- At 1M min/month → **~\$84K/month** saved on LLM inference alone
- Self-hosted ROCm TTS removes **\$100K–\$300K/month** hosted synthesis
- India on-prem: **~50% lower GPU CapEx**

The advantage is **real** — yet unrealized, because AMD is absent from the ecosystem surface where compute is **selected**.

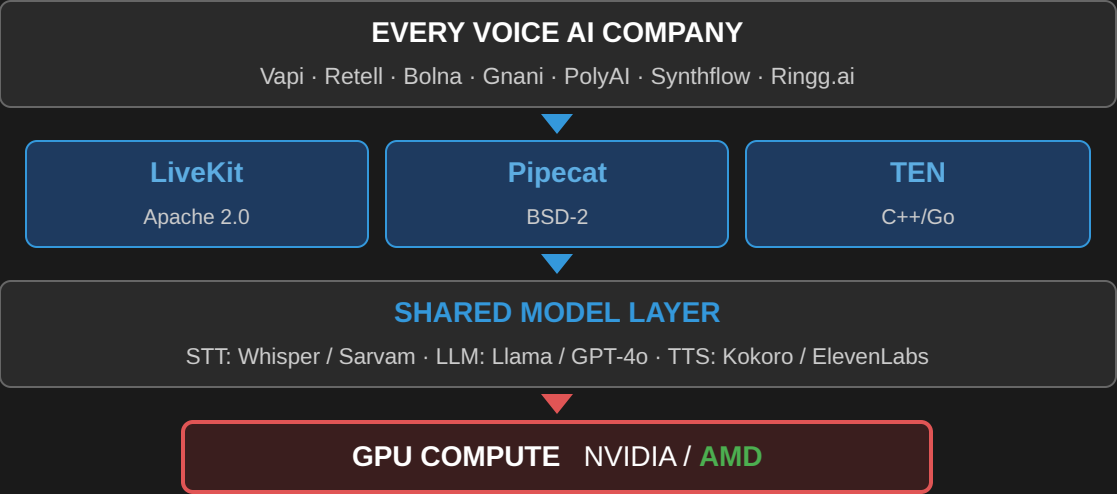
The Voice AI Stack and Its Convergence

Fig 1 · Five-layer pipeline — 3 layers are GPU-compute-bearing (*median E2E 680 ms; bar → sub-500 ms*)



Shaded **ASR · LLM · TTS** consume GPU cycles — AMD's advantage *and* its gaps both live here.

Fig 2 · Industry convergence — every platform sits above one of three frameworks



Win the **compute layer** — or the **plugin slot just above it** — and the hardware choice is made for every platform above.

CUDA Gap Closed; AMD Still Lacks a Low-Resistance Path to Its Silicon

Table 1 · ROCm support for voice-AI components (2026)

Component	ROCm	Note
LLM serving (vLLM, Llama 3.1)	Full	90–95% of H100 tput
LLM serving (SGLang)	Full	Within 5–10% of CUDA
TTS: Kokoro, Chatterbox	Full	Pure PyTorch, 0 changes
TTS: Coqui XTTS-v2	Partial	Works with effort
ASR: Faster-Whisper (batch)	Full	RT mode needs work
ASR: streaming containers	Gap	CUDA-assumed
Embeddings: HF TEI	Gap	No ROCm build

Two heaviest layers (LLM, TTS) solved; streaming ASR + TEI remain open.

Table 3 · NIM ecosystem gap: NVIDIA vs AMD required

NVIDIA Has	AMD Equivalent	Status
Riva ASR NIM	AMD ASR container	Absent
Riva TTS NIM	AMD TTS container	Absent
AI Blueprint: voice agent	Pipecat on MI300X	Absent
Pipecat NVIDIA processor	Pipecat AMD processor	Absent
LiveKit certified backend	LiveKit AMD backend	Absent
VS Code NIM extension	ROCm / Quark ext.	Absent
Azure Foundry NIM	Azure AMD endpoint	Partial
SI cert (TCS, Infosys)	SI cert program	Absent
28M devs default CUDA	AMD dev default	Opt-in

Table 4 · Framework ecosystem presence

Framework	NVIDIA	AMD
Pipecat	Official processor	Absent
LiveKit Agents	Certified backend	Absent
LangChain	langchain-nvidia-ai-endpoints	Absent
LlamaIndex	llama-index-llms-nvidia	Absent
Haystack	NIM pipeline component	Absent

The gap is not silicon — it's the least-resistant path to silicon: containers, plugins, blueprints & marketplace slots.

The Five Downstream Ecosystem Layers

Same logic at every layer: ship a named package → developers adopt the default → the hardware choice becomes invisible at the call site.

1 · Framework plugin slots

`ChatNVIDIA(...)` routes to NVIDIA at the endpoint. AMD doesn't exist at that call site — repeated across millions of pipelines.

2 · Inference serving infrastructure

Triton / Dynamo + TensorRT-LLM = the invisible substrate (20–30% edge at batch 1–4). AMD has **vLLM ROCm only** — one tool, not infrastructure.

3 · Voice-domain SDKs

Riva (ASR+TTS+MT+diarization) · Maxine (noise suppression) · ACE (digital humans). **AMD has no equivalent for any of the three.**

4 · Edge hardware platform

Jetson (Nano → Thor) + JetPack ships Riva, vLLM, CUDA. Certified OEMs: Siemens, KUKA, ADLINK. Caterpillar runs Jetson Thor + Riva inside a bulldozer cab. AMD Ryzen AI = **laptop only** — no industrial module.

5 · Cloud marketplace slots

NIM native on Azure Foundry; selectable on HF, SageMaker, Vertex. AMD **absent / limited** — TGI ROCm image is production-ready but **not surfaced in the UI**.

"Deploy this model" → MI300X isn't in the GPU dropdown. Not a silicon gap — a **slot gap**. And a slot gap is **closeable**.

The Four Upstream Gaps

Gap 1 · Training & fine-tuning moat

Voice companies fine-tune weekly (custom ASR/TTS/NLU). **NeMo** → **Triton export** locks the whole loop to NVIDIA. Key libs are CUDA-primary on AMD:

Library	Role	ROCm Status
<code>bitsandbytes</code>	4/8-bit quant	Community fork
<code>trl</code> (HF)	RLHF / fine-tune	Experimental
Axolotl	Most-used tuner	Untested
PEFT	LoRA / QLoRA	Works, less tested
FlashAttention 3	Fastest attn kernel	No AMD equiv.

Gap 2 · ISV & SI sales channel

GPU chosen implicitly by the ISV. NVIDIA is in **Genesys** (11k+ customers), **Cisco Webex** (Maxine), **Salesforce**, **ServiceNow**. SIs (TCS, Infosys, Wipro, Accenture) are **certified on NVIDIA**. AMD: no presence in any major contact-center ISV.

Gap 3 · Voice + vision convergence

Next-gen agents fuse camera + mic. Jetson Thor unifies **Isaac** · **Metropolis** · **Riva** · **ACE** on one substrate. AMD's lines don't converge (MI300X · Ryzen AI · Radeon) — a **missing product category**.

Gap 4 · Synthetic data for low-resource languages

No AMD-native speech-data pipeline or academic partnership (e.g. AI4Bharat) → lifecycle dependency predates any inference decision.

Closing **downstream** puts AMD in contention; closing **upstream** creates **platform momentum**. The chain compounds: fine-tune on NeMo → deploy to Triton → shipped inside Genesys → sold by an NVIDIA-certified SI.

The AMD Gateway Strategy

Run NVIDIA's playbook — targeting the cost wedge its pricing creates. Same Pipecat code, same LiveKit transport, 34% lower per-conversation LLM cost.

Step 1 · Three certified containers

`docker pull` → validated in 30 s: (a) Pipecat Voice Agent (Whisper + vLLM + Kokoro), (b) LiveKit + SIP, (c) Sarvam India (DPDP-compliant, on-prem).

Step 2 · AMD Voice AI Blueprints

Deployable repos, not docs — US/EU contact center, India BFSI, Arabic GCC, edge/offline. Must run on a **~\$700 consumer GPU**.

Step 3 · Pipecat ROCm plugin

`pipecat-ai[amd]` on PyPI — AMD as an official processor alongside Deepgram / ElevenLabs. **~2 weeks**.

Step 4 · HF TEI on ROCm

Upstream ROCm into Text-Embeddings-Inference — closes the last RAG software gap.

Step 5 · HF Inference Endpoints slot

TGI ROCm is ready — one **business agreement** makes MI300X selectable. **Zero eng weeks**.

Step 6 · Ryzen AI NPU on-ramp

`pipecat create --backend amd-npu` — offline, no GPU, no API key. The dev-laptop demo that mirrors **RTX Spark**.

PAVS sits exactly at this intersection — between AIG model outputs and lighthouse customers. Converting a raw model → production vertical software does *organically* what NVIDIA's platform teams do *intentionally*. The pattern generalizes: robotics, industrial, healthcare, automotive.

Prioritized Action Roadmap

Tier 1 — Pure engineering (no new silicon, no product decisions)

Action	Effort	Horizon	Impact
HF Inference Endpoints slot	Business neg.	0–4 wks	High
Certified ASR + TTS containers	2 wks each	1 month	High
Pipecat AMD processor (PyPI)	2 wks	1 month	Medium
LangChain / LlamaIndex / Haystack	2–4 wks ea.	1–2 mo	High
AMD Voice AI Blueprints (4 targets)	4 wks each	2 months	High
bitsandbytes ROCm official	4–6 wks	2 months	High
Triton ROCm certified image	12–16 wks	4 months	High
Ryzen AI NPU voice template	4 wks	2 months	Medium

Tier 2 — Business dev + engineering

Action	Effort	Horizon	Impact
Genesys ISV integration	Business dev	6–12 mo	Very high
SI cert program (TCS, Infosys)	Program build	6–12 mo	Very high
FlashAttention 3 / CDNA3 equiv.	16–24 wks	6 months	Medium

Tier 3 — Strategic HW commitments

Action	Effort	Horizon	Impact
AMD voice data initiative (India)	\$5–13M, 2 yr	18–24 mo	Strategic
Jetson-equiv. industrial edge module	HW roadmap	2–3 yrs	Very high

First 12–16 weeks: Steps 1–4 with 6–8 engineers make AMD visible to every Pipecat & LiveKit developer at first contact. **India BFSI under DPDP** is the natural beachhead — on-prem mandatory, ~50% lower CapEx.

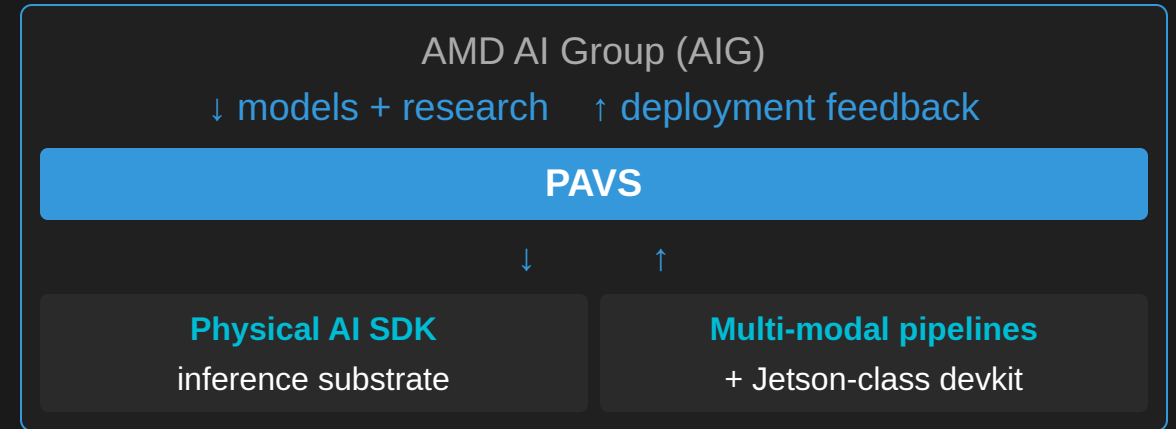
How the Bridge Is Being Closed

The gap can't be closed by a new chip

- It's a **software & ecosystem** problem — containers, plugins, blueprints, marketplace slots
- Which is **exactly** the problem a **vertical-software team** solves

PAVS produces the closing assets organically

- Each lighthouse deployment → a **reusable pipeline** = a step toward an AMD-native ecosystem
- Each model the Physical AI SDK supports → **one less** requiring NVIDIA tooling on AMD HW
- Each robotics pipeline on Ryzen AI APU → a **reference design** for an OEM / SI / ISV



The window is open — but NVIDIA's Jetson ecosystem, NIM integrations & SI certs deepen every quarter. **Make AMD the path of least resistance at the first touchpoint, and the rest of the stack follows adoption.**